



Theoretical ecotoxicology

Multiple stressors: Stressor interactions

My research

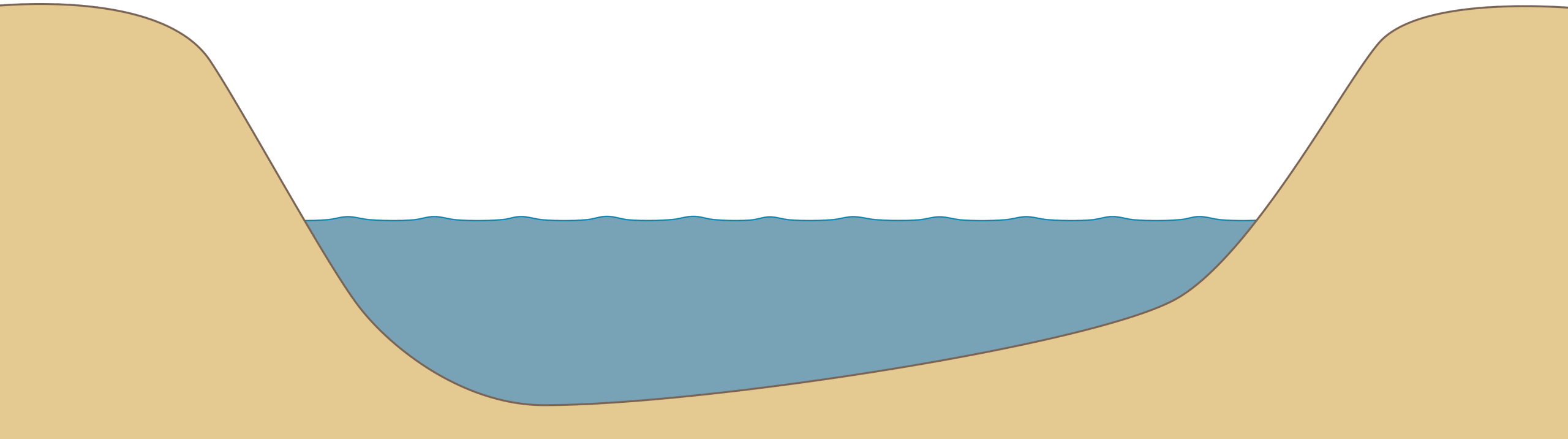
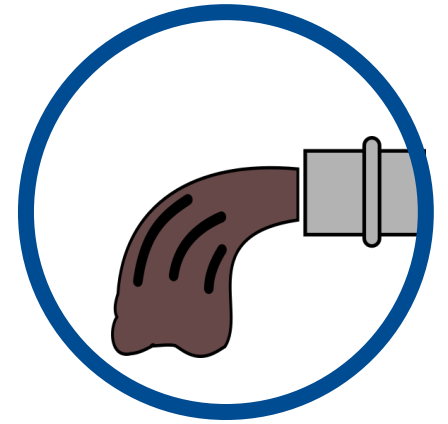
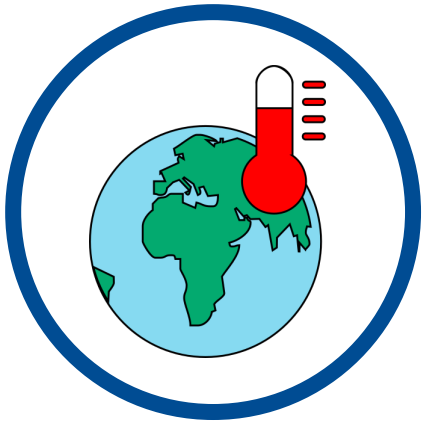


- Doctoral researcher
- Multiple stressor effects on macroinvertebrate communities
 - DNA-based methods
 - Mesocosm experiments
 - Theoretical work on how multiple stressors combine



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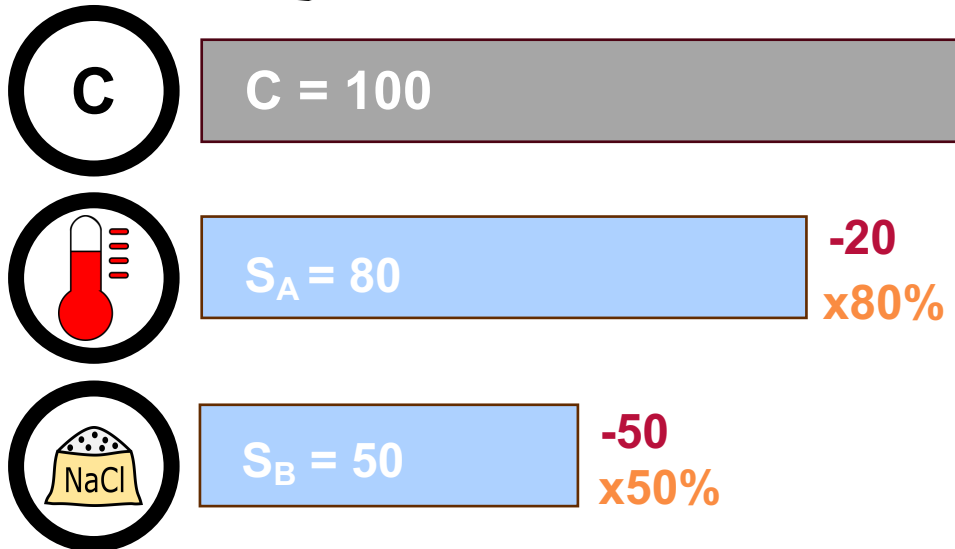
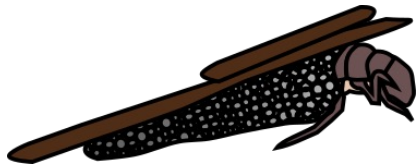
Streams are threatened by many stressors



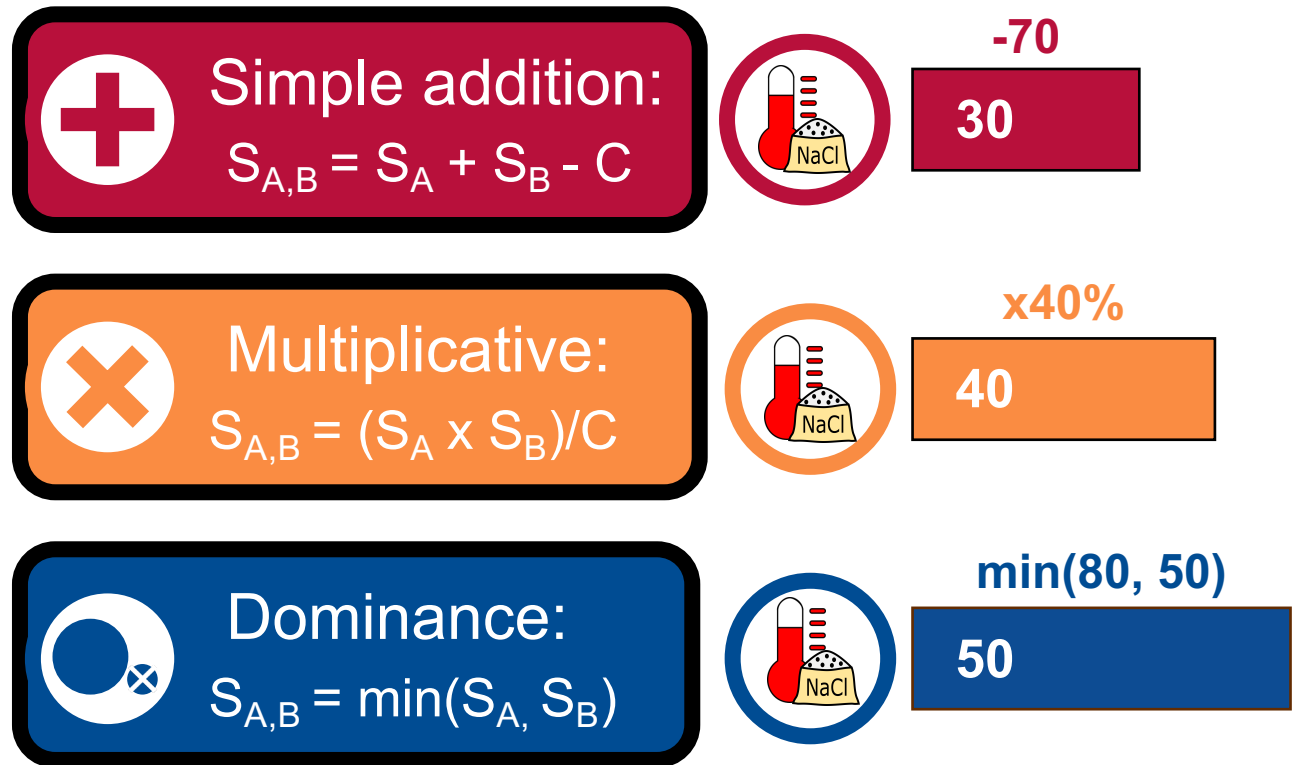
Goal: Predict how two or more stressors combine

Multiple-stressor null model:

A model that predicts the joint effect assuming the absence of interactions



Null models for stressor co-occurrence

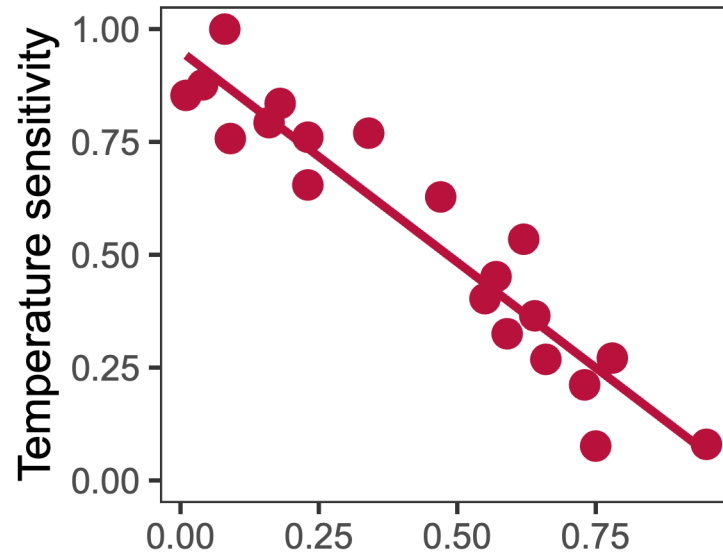


If stressor effects are exclusively negative

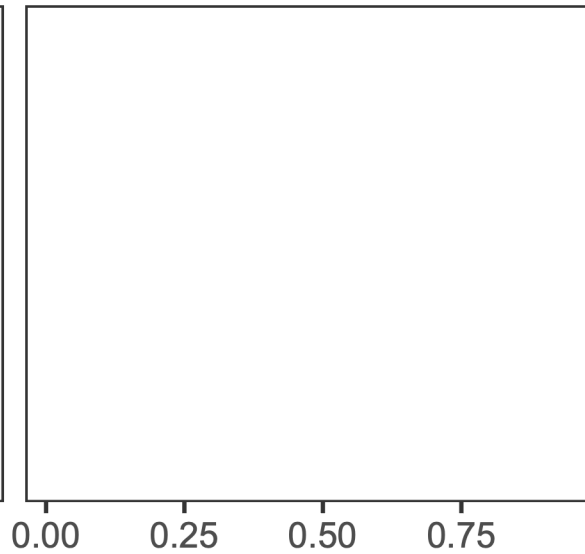
Which null model should I choose?

Stressor sensitivity correlation

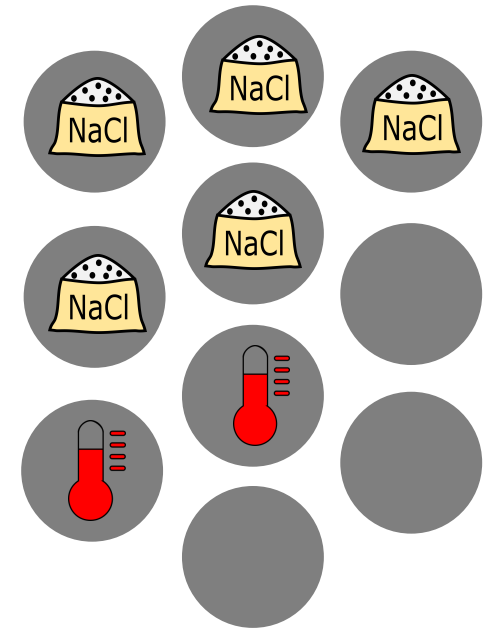
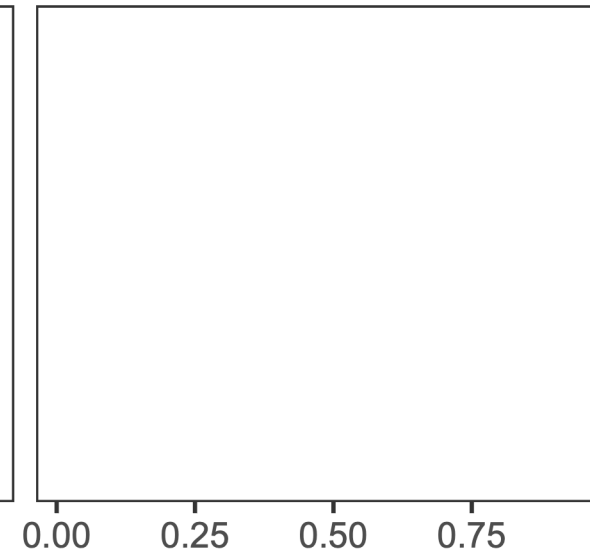
Simple addition



Multiplicative



Dominance

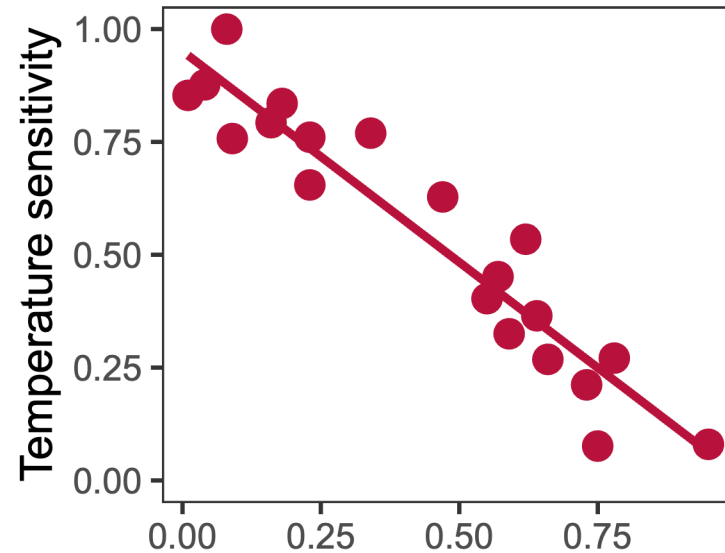


Negative correlation:
Each stressor affects a
different set of individuals
(or species)

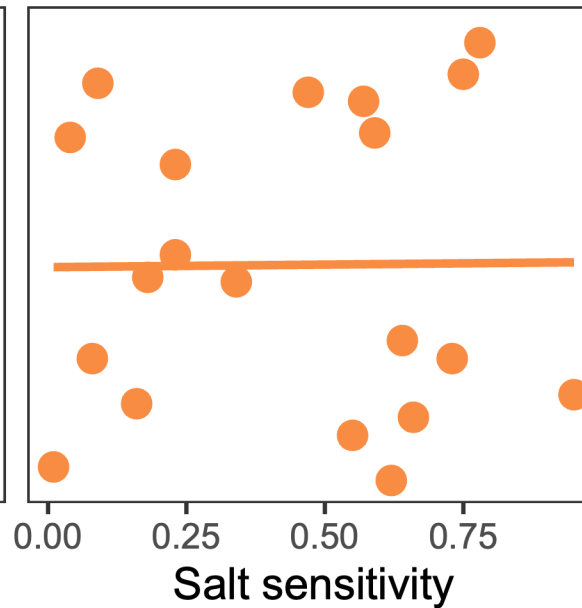
Which null model should I choose?

Stressor sensitivity correlation

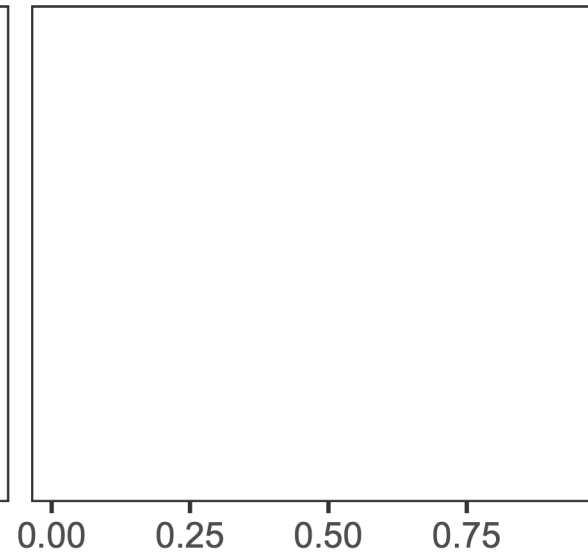
Simple addition



Multiplicative

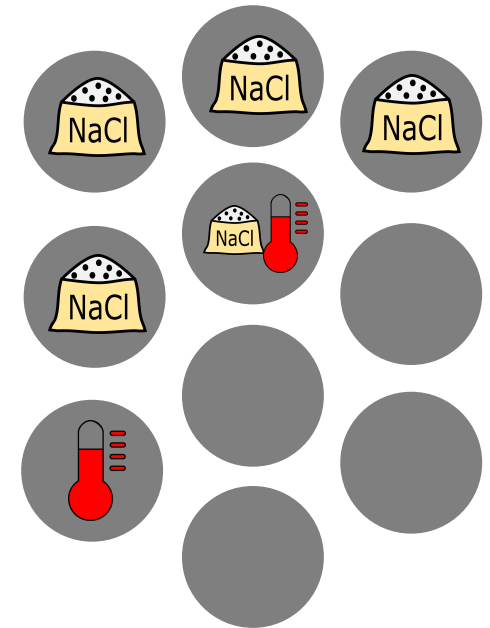


Dominance



Negative correlation:
Each stressor affects a
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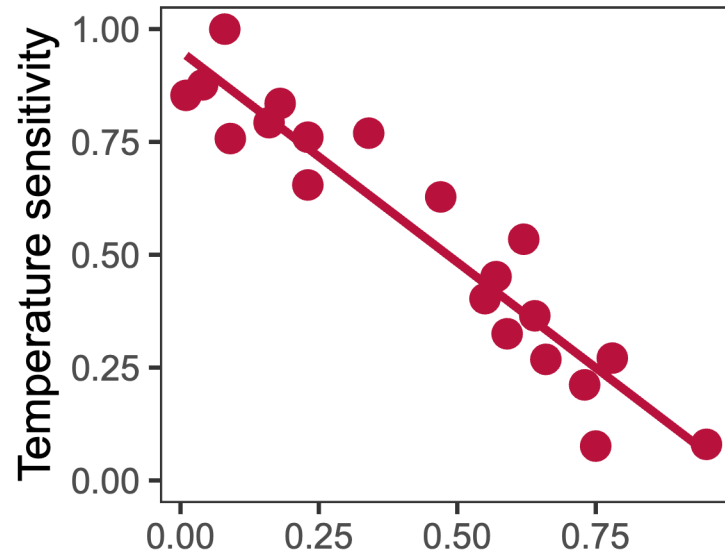
No correlation:
Stressor sensitivities are
independent of each
other



Which null model should I choose?

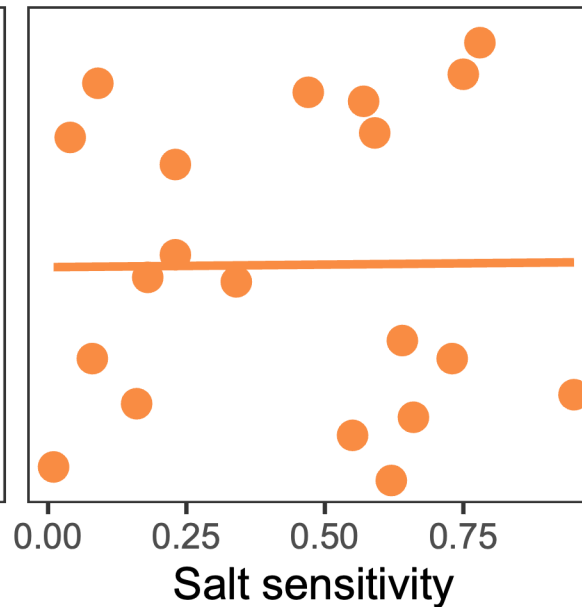
Stressor sensitivity correlation

Simple addition



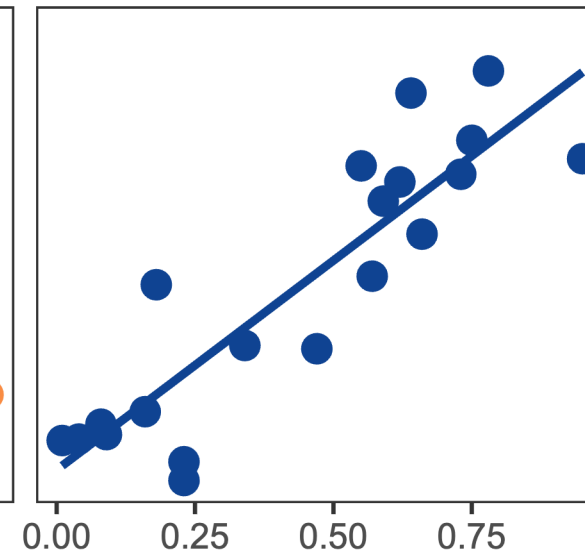
Negative correlation:
Each stressor affects a different set of individuals (or species)

Multiplicative

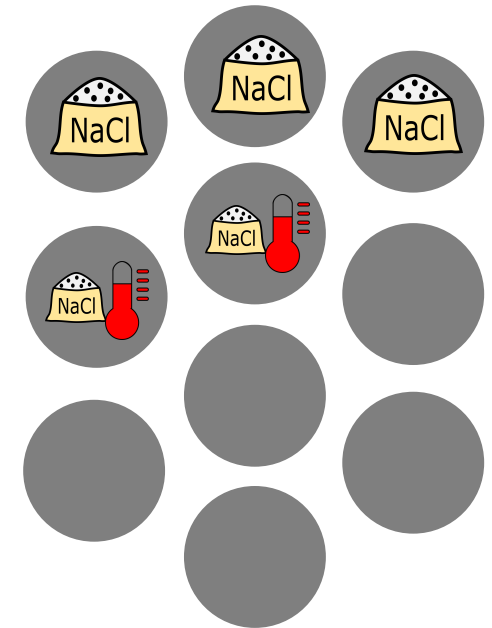


No correlation:
Stressor sensitivities are independent of each other

Dominance



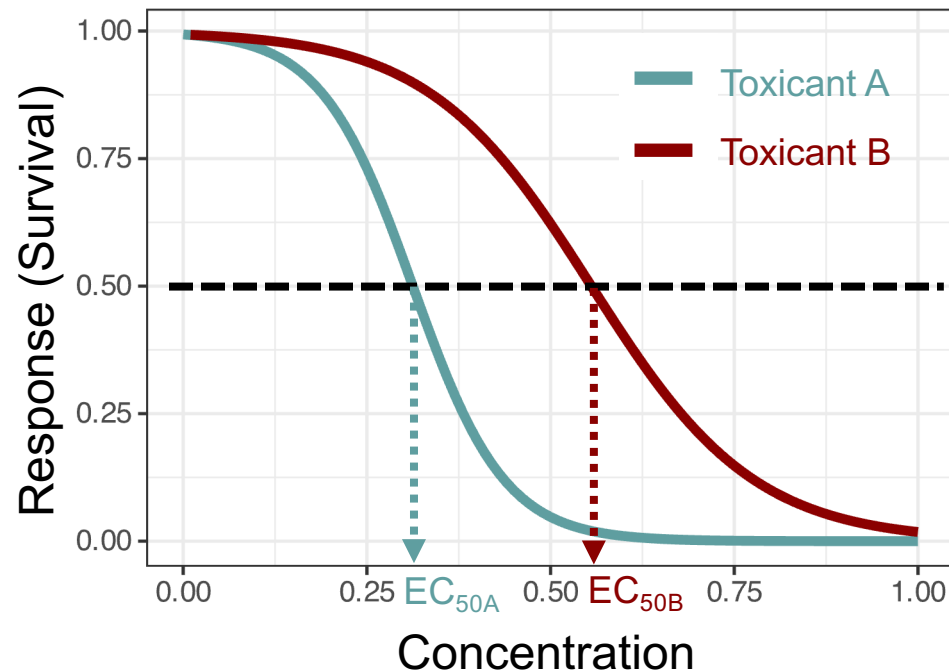
Positive correlation:
If an individual (or species) is affected by one stressor, it is likely to be also affected by the other



Toxicant mixtures: Null models in ecotoxicology

Similar modes of action: Concentration addition

$$f_{A,B}(c_A, c_B) = f_A(c_A + \lambda c_B) \quad \text{Potency } \lambda = \frac{EC_{xA}}{EC_{xB}}$$



Different modes of action: Independent action

$$E_{A,B} = E_A + E_B - E_A E_B$$

$$\downarrow E_X = 1 - \frac{S_X}{C}$$

$$1 - \frac{S_{A,B}}{C} = 1 - \frac{S_A}{C} + 1 - \frac{S_B}{C} - \left(1 - \frac{S_A}{C}\right) \left(1 - \frac{S_B}{C}\right)$$

$$\downarrow \text{Simplification}$$

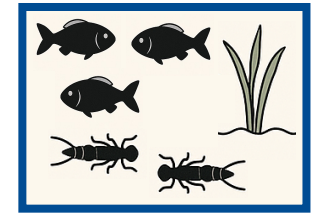
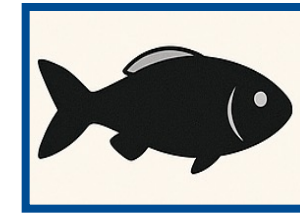
$$S_{A,B} = \frac{S_A S_B}{C}$$

Mechanisms of stressor interactions

How do stressor interactions manifest?

- **Intensity interactions**
 - Changes in stressor intensity (or exposure) by another stressor
 - E.g.: Drought increases pollutant concentrations in streams
- **Effect interactions**
 - Changes in stressor effects by another stressor
 - E.g.: Higher toxicity of a pesticide if oxygen supply is low

Where do stressor interactions occur?

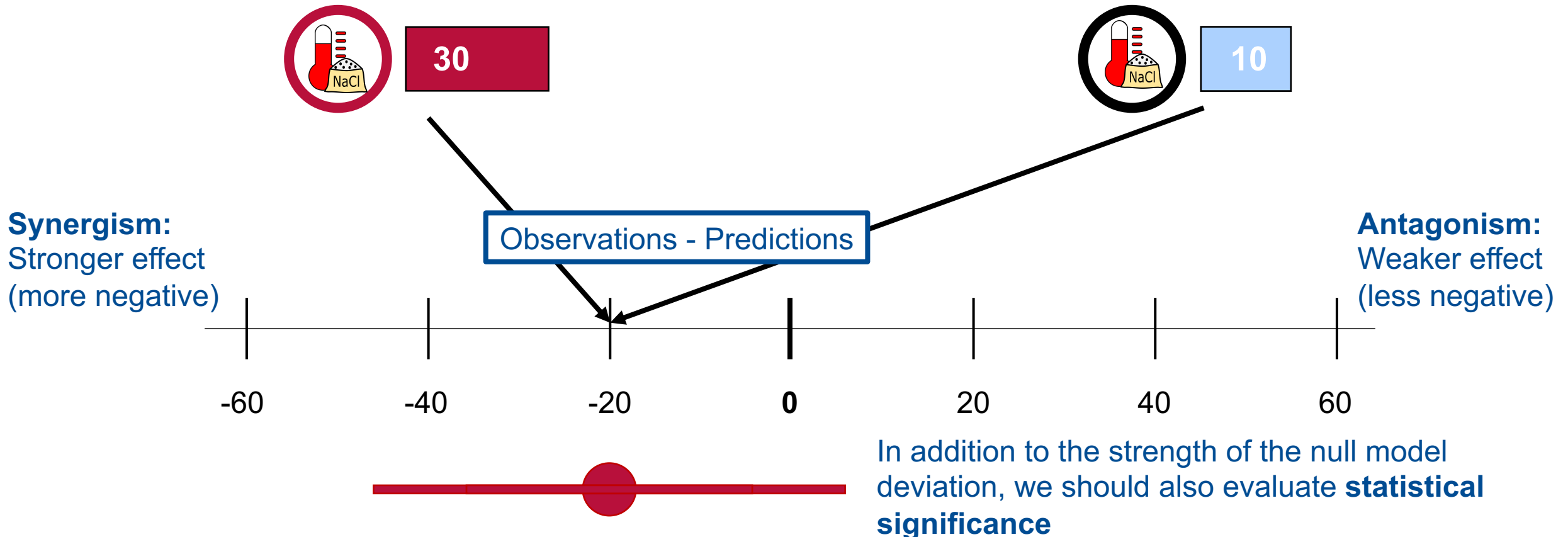


- **Abiotic environment**
 - Purely physical/chemical processes
- **Organismal level**
 - Through physiological processes within an organism
- **Ecosystem level**
 - Alterations of biotic interactions

How can we test for stressor interactions?

Null model predictions
(Simple addition)

Experimental observations



Interaction terms in multiple linear regression

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

Intercept

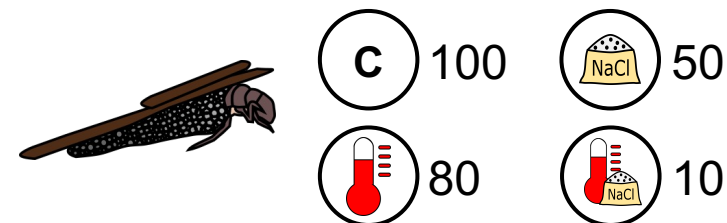
What is the response if both predictors are absent (at value 0)?

Predictor main effects

How much does the response change for one unit increase in x_1 (or x_2)?

Interaction effect

How much does the per-unit effect of x_1 change with one unit increase in x_2 (and vice versa)?

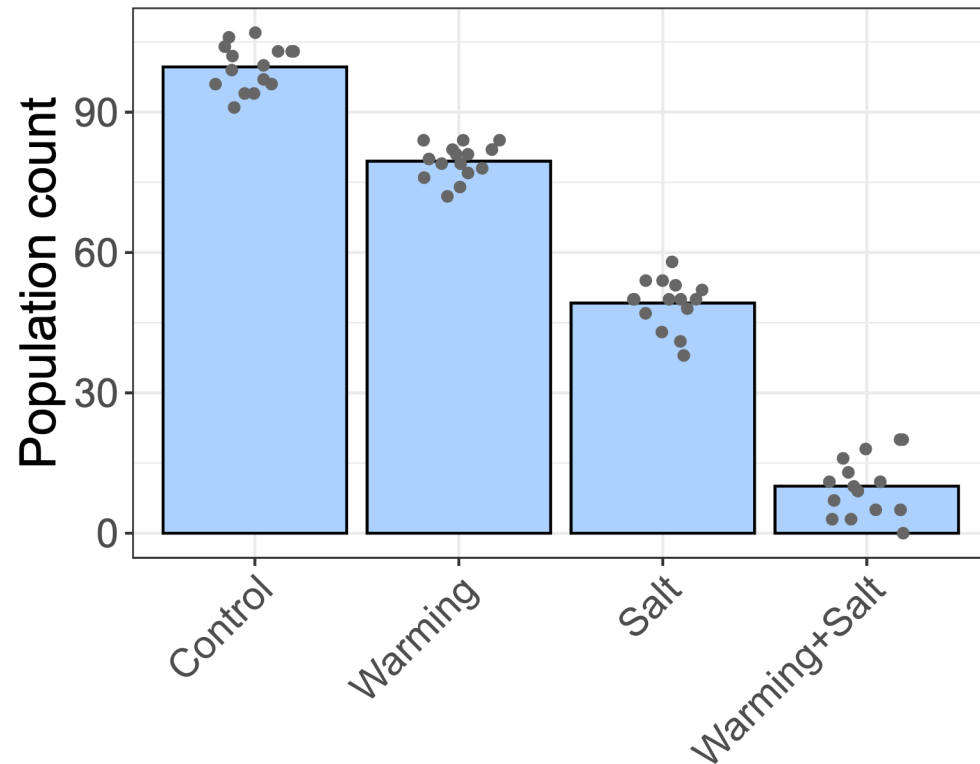


$$y = 100 - 20x_1 - 50x_2 - 20x_1x_2$$

Linear regression evaluates the simple addition null model

Interactions between categorical predictors

By how much does the effect of salt change if warming is also present? (And vice versa)

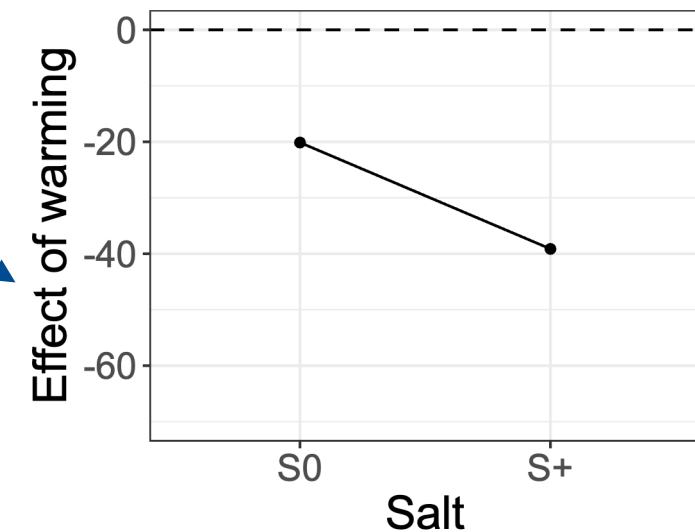
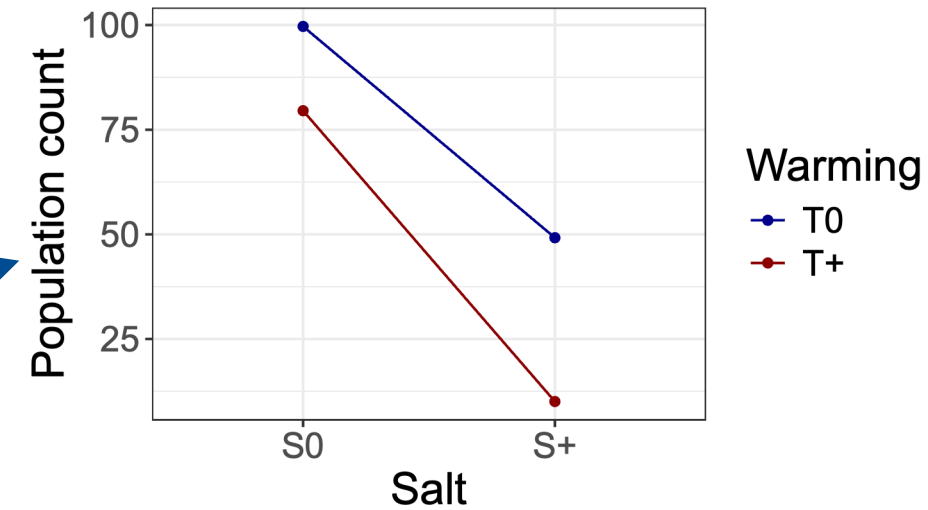


Null model:
Parallel lines

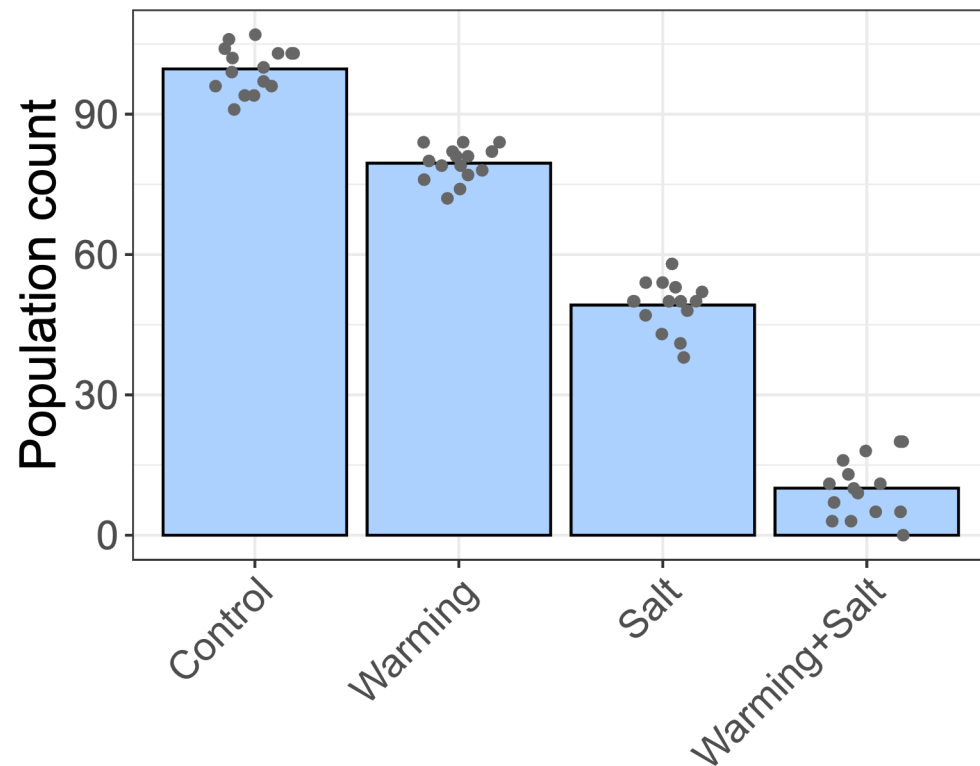
Conditional plot

Marginal effects plot

Null model:
Horizontal line



Interactions between categorical predictors

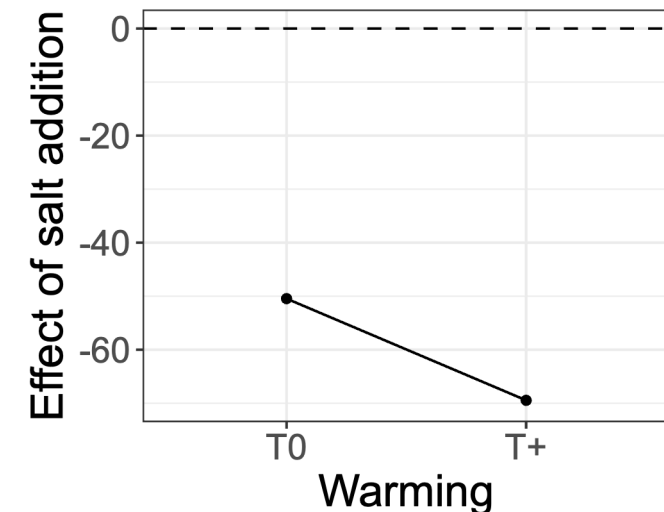
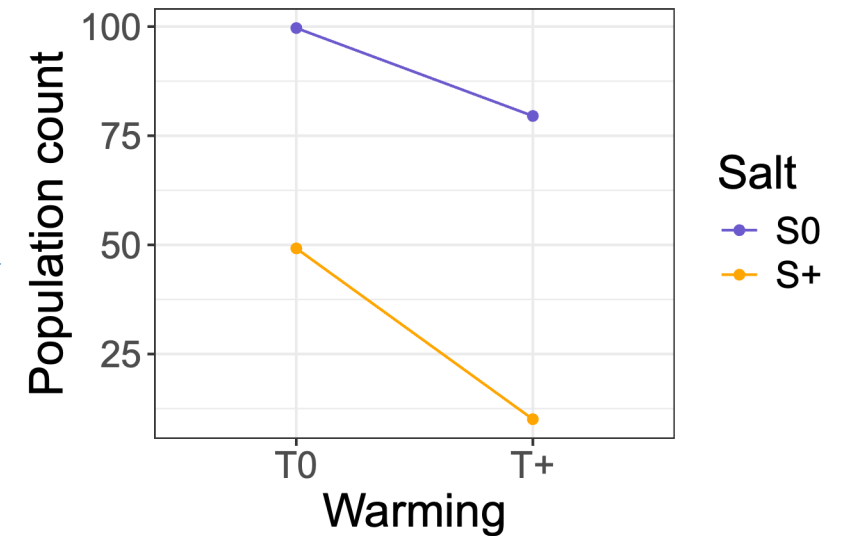


Null model:
Parallel lines

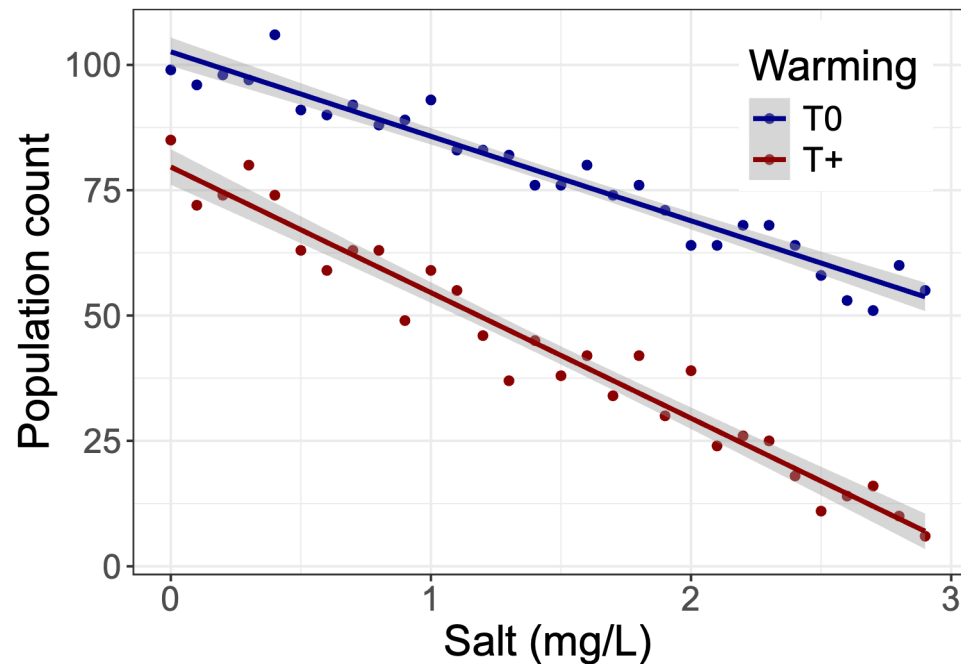
Conditional plot

Marginal effects plot

Null model:
Horizontal line

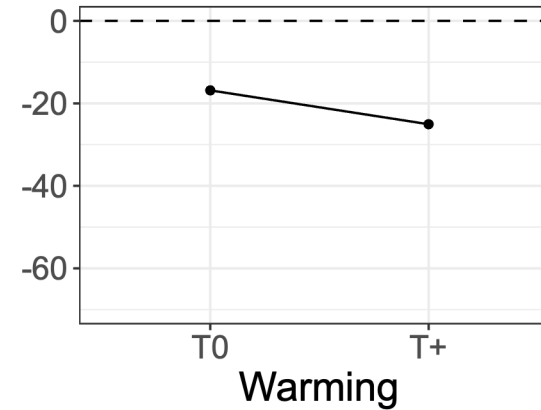


Interaction between a categorical and continuous predictor



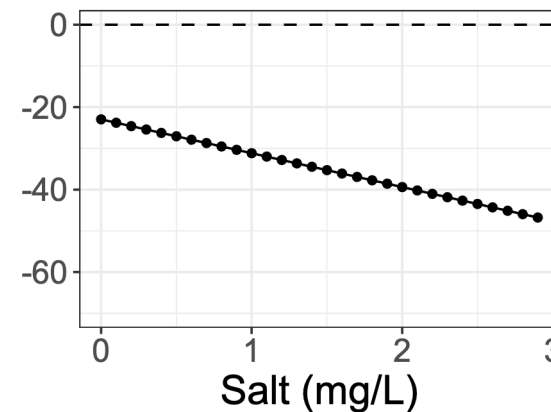
Marginal effects plot

Slope for salt



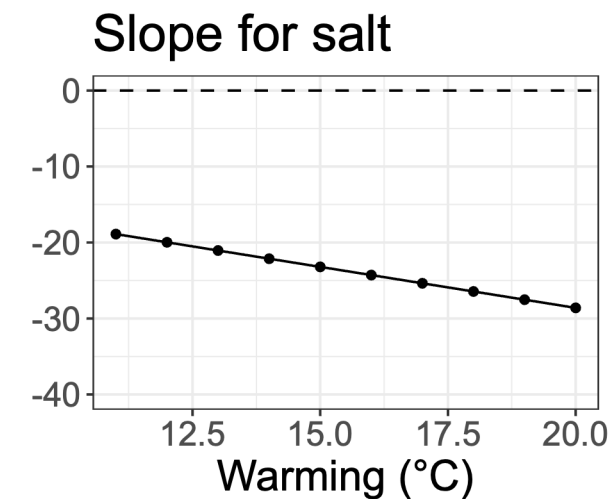
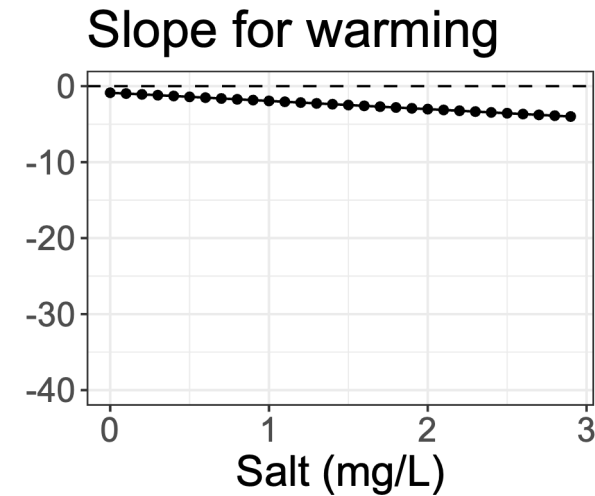
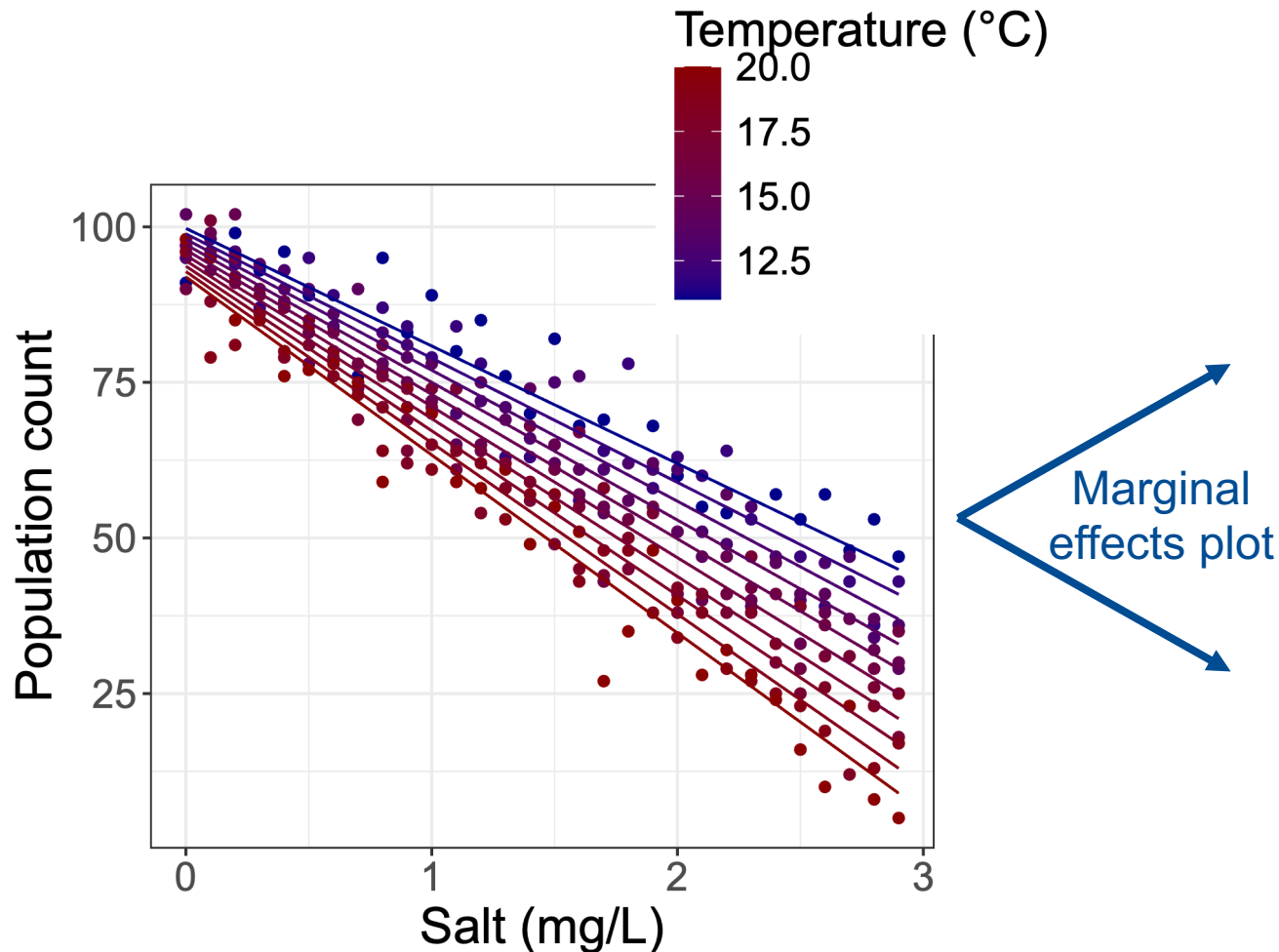
Describes how the slope of the salt regression line changes between the two temperature conditions

Effect of warming



Describes how the difference in population size between control and warming changes with increasing salinity

Interaction between two continuous predictors



Describes how the slope of the regression line for stressor A changes for a unit increase in stressor B

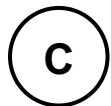
Interpretation of model output

2 categorical predictors

Coefficients:

	Estimate	Std. Error
(Intercept)	99.667	1.321
temperatureT+	-20.133	1.869
salinityS+	-50.467	1.869
temperatureT+:salinityS+	-19.000	2.643

$$y = 99.7 - 20.1 \times T - 50.5 \times S - 19 \times T \times S$$



$$T = 0, S = 0$$



$$T = 0, S = 1$$



$$T = 1, S = 0$$



$$T = 1, S = 1$$

2 continuous predictors

Coefficients:

	Estimate	Std. Error
(Intercept)	108.6765	3.2143
temperature	-0.8525	0.2039
salinity	-7.2802	1.9034
temperature:salinity	-1.0395	0.1207

$$y = 108.7 - \frac{0.9 \times T}{^{\circ}\text{C}} - \frac{7.3 \times S}{\text{mg/L}} - \frac{1 \times T \times S}{^{\circ}\text{C} \times \text{mg/L}}$$

Slope of
temperature for
0 mg/L of salt

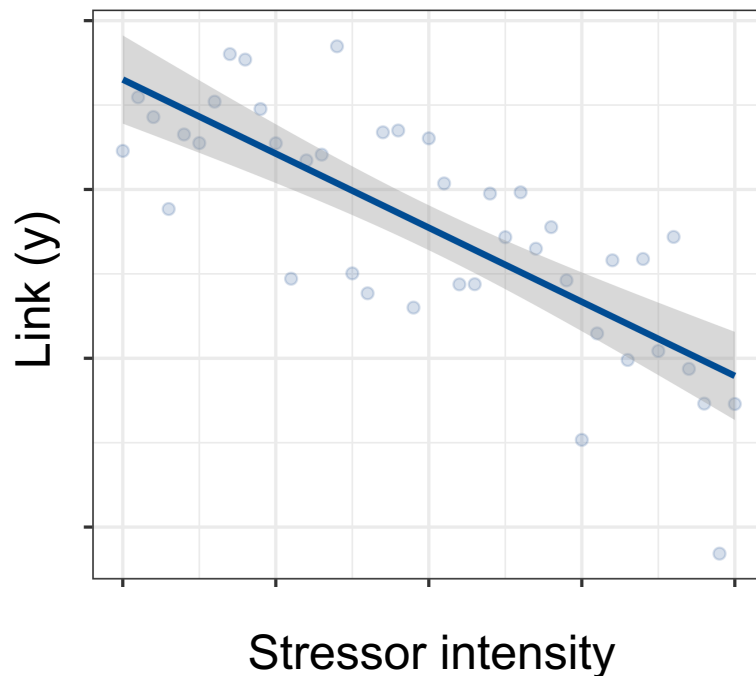
Slope of salinity
for 0 °C

Interactions in Generalized linear models: Link scale vs. Response scale

Response variable
and link function

$$\text{link}(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

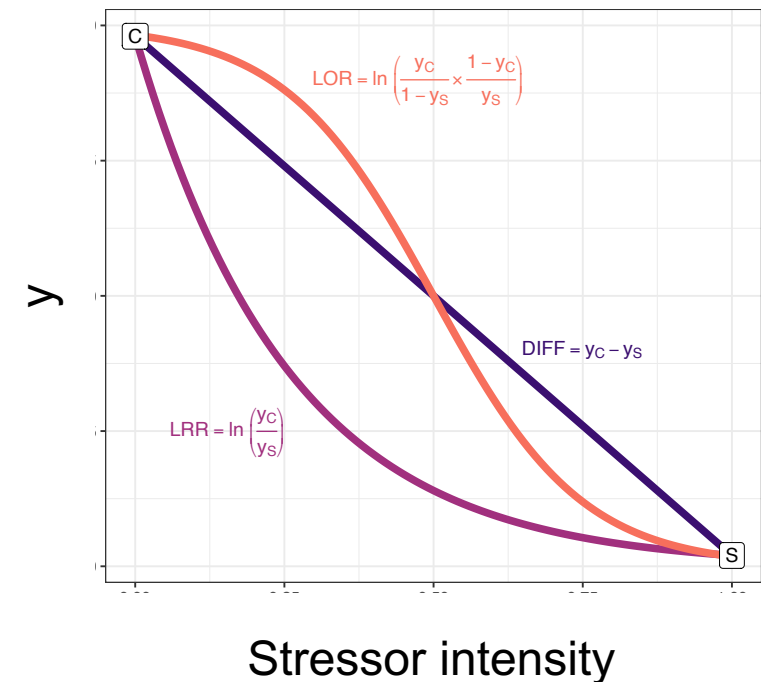
Link function linearizes
stressor-response
relationships



Identity link
(Gaussian regression)

Logarithmic link
(Poisson regression)

Logit link
(Binomial regression)



GLMs have a canonical link function

GLM	Used for...	Limits of response	Canonical link
Gaussian	Normal data	$(-\infty, +\infty)$	Identity
Poisson	Count data	$(0, +\infty)$	Logarithmic
Binomial	Proportional data	$(0, 1)$	Logit
Gamma	Continuous data ≥ 0	$(0, +\infty)$	Inverse

- Canonical link function facilitates the estimation process for the model parameters
- Most typically used link function for a GLM
- Other link functions can also be used, but this may lead to convergence issues

What does the (canonical) link function do?

It determines...

- 1. The stressor-response relationship that is modelled**
- 2. The effect measure**
 - Differences (identity link)
 - Ratios (logarithmic link)
 - Odds ratios (logit link)
- 3. How the predictors' effects are expected to combine in the absence of interactions**

How the logarithmic link implies the multiplicative null model

Absence of
interaction: $\beta_3 = 0$

Apply inverse link
function (here:
exponentiate)

Separate
predictor effects
according to
algebraic rules

$$\log(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$



$$\log(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$



$$y = e^{\beta_0 + \beta_1 x_1 + \beta_2 x_2}$$



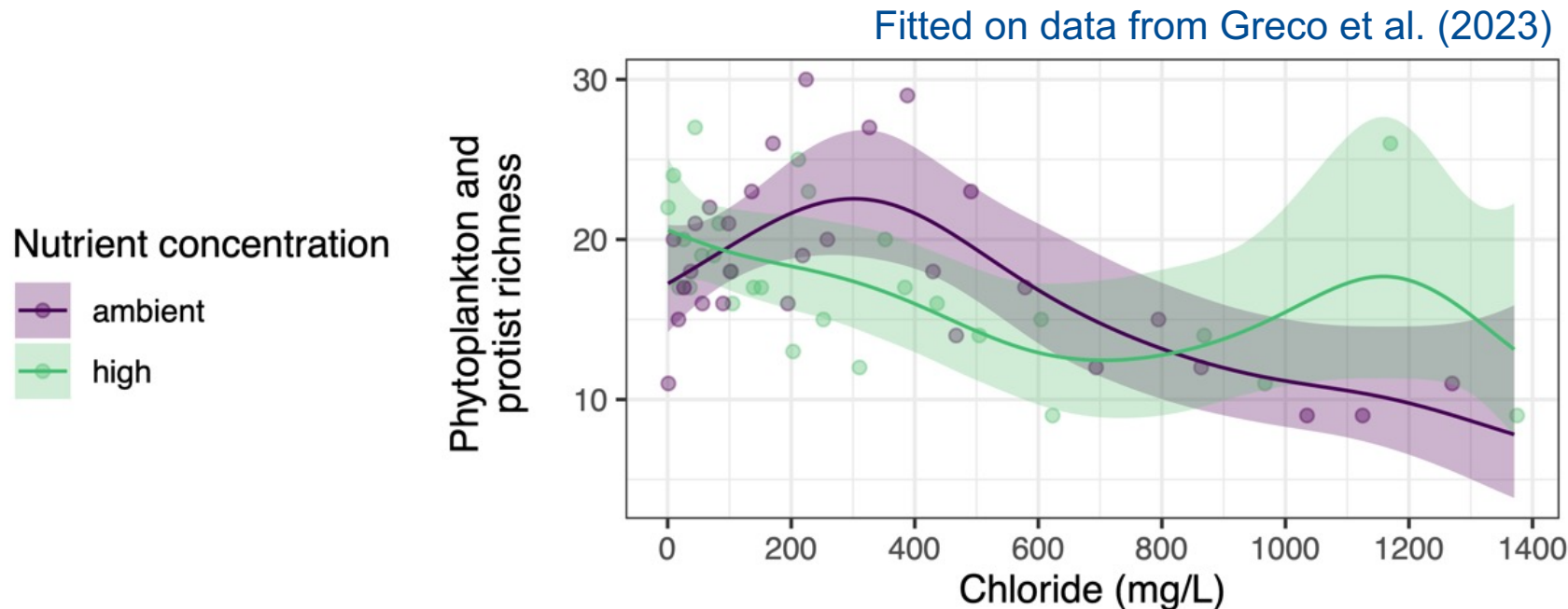
$$y = e^{\beta_0} \times e^{\beta_1 x_1} \times e^{\beta_2 x_2}$$

The logarithmic
link function in a
GLM evaluates
the multiplicative
null model

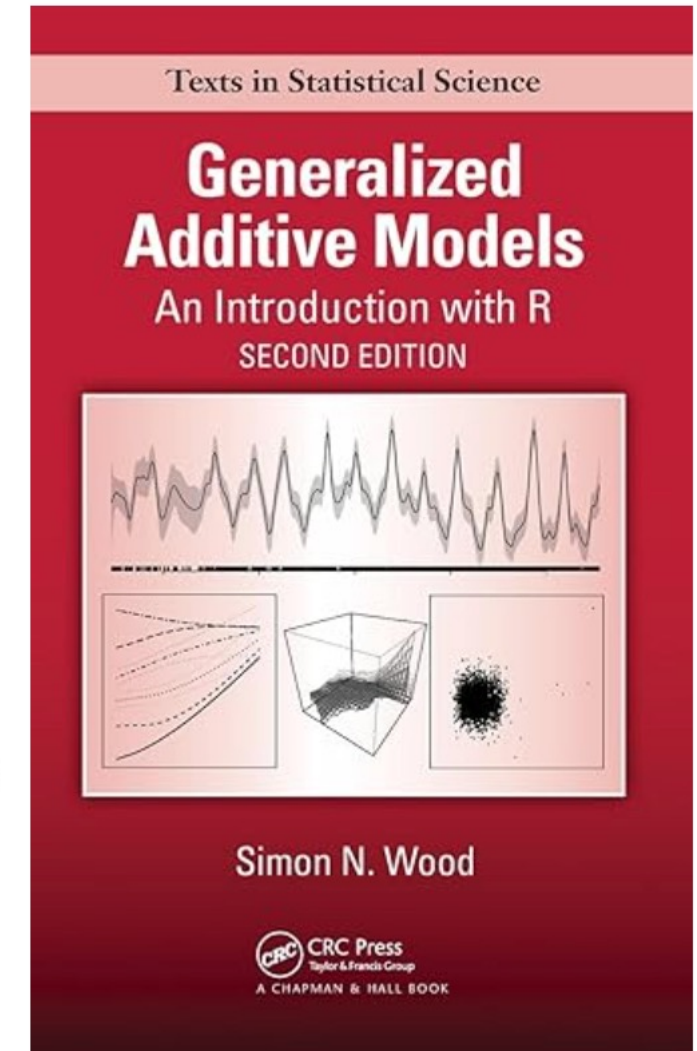
1. **Stressor interactions are deviations from null model predictions**
2. **Null model choice should be based on *a priori* knowledge on...**
 - Similarity in stressor/toxicant mode of action
 - Sensitivity correlations
3. **GLMs make their own assumption on how the effects of predictors combine in the absence of an interaction**
 - Identity link: Simple addition null model
 - Logarithmic link: Multiplicative null model

Outlook: Non-linear relationships are common in ecology and require GAMs

Generalized additive models use smoothers to model non-monotonic functions

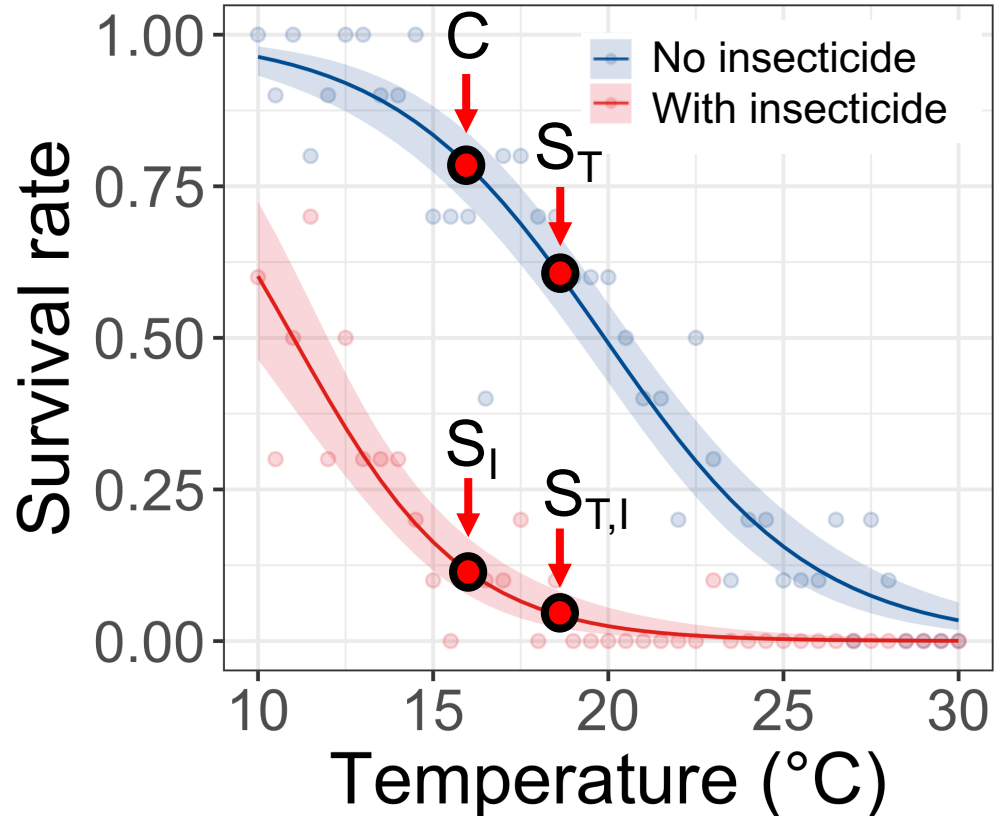


In R: *mgcv* package



Outlook: How can we test our null model of choice?

(1) Fitted and validated regression model (logit link)



(2) Post-estimation hypothesis testing for null model evaluation

$$S_{T,I} = \frac{S_T \times S_I}{C}$$

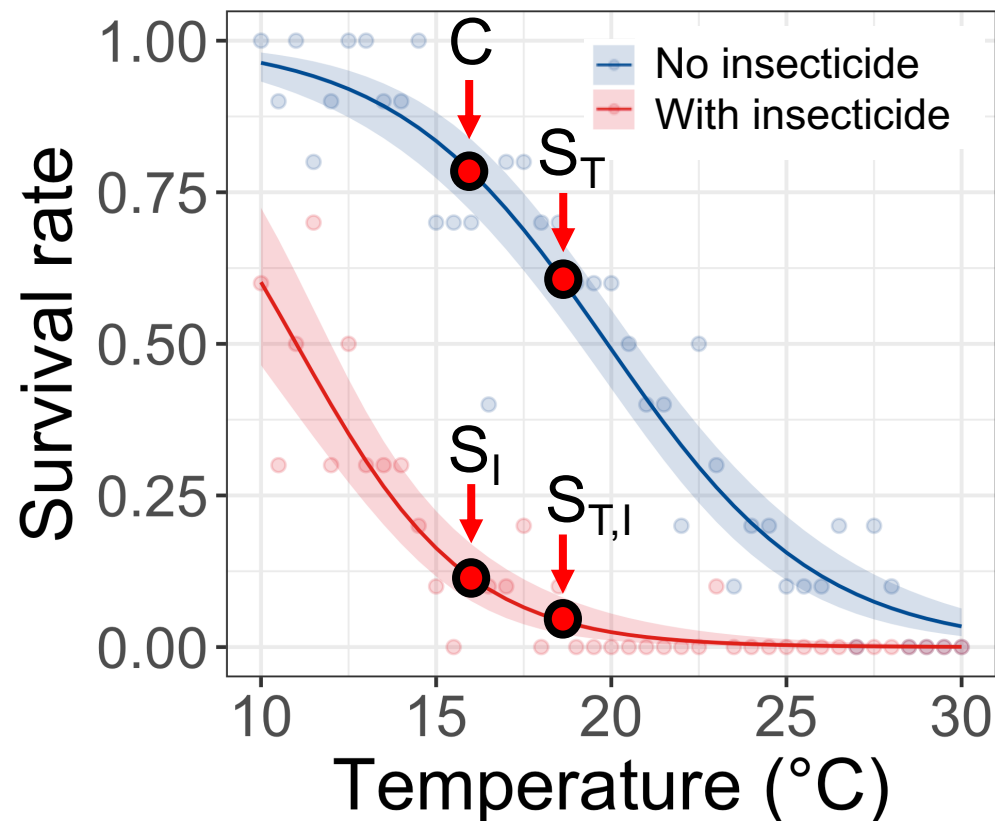
↓

$$Int_{MU} := \log \left(\frac{S_{T,I} \times C}{S_T \times S_I} \right)$$

If null model is true:
 $Int_{MU} = 0$

Outlook: How can we test our null model of choice?

(1) Fitted and validated regression model (logit link)



(2) Post-estimation hypothesis testing for null model evaluation

